

1.2 Atoms, Molecules & Stoichiometry

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.2 Atoms, Molecules & Stoichiometry
Difficulty	Medium

Time allowed: 70

Score: /53

Percentage: /100

Question 1a

This question is about the formation and reaction of silane, SiH_4 .

0.5 g of quartz sand and 0.5 g of powdered silica are mixed with 2.4 g of magnesium powder. The resulting mixture is placed in a pot half filled with sand. A magnesium ribbon fuse is added and the mixture is ignited to form magnesium oxide and solid magnesium silicide, Mg_2Si .

i)

Write an equation for this reaction.

Include state symbols.

[2]

ii)

Determine the limiting reagent in this reaction.

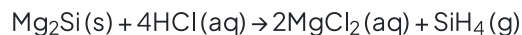
Show your working.

[2]

[4 marks]

Question 1b

The magnesium silicide is reacted with hydrochloric acid according to the following equation:



The gaseous silane formed spontaneously ignites in air.

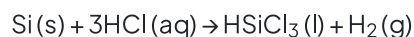
State the type of reaction and write the chemical equation for this reaction.

Include state symbols.

[3 marks]

Question 1c

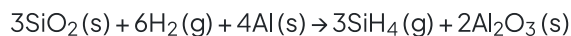
An alternative method to produce silane involves the reaction of silicon with hydrochloric acid to form trichlorosilane, HSiCl_3 .



Calculate the maximum mass of trichlorosilane produced from the reaction of 35.125 g of silicon.

[3 marks]**Question 1d**

Another method of silane production involves the reduction of silicon dioxide by aluminium and hydrogen gas.



What is the volume, in dm^3 , of hydrogen gas required to produce 10 dm^3 of silane gas? Explain your answer.

[2 marks]**Question 2a**

Cobalt(II) sulfate is used in storage batteries, electroplating baths and as an additive to soils and animal feeds.

A sample of hydrated cobalt sulfate, $\text{CoSO}_4 \cdot x\text{H}_2\text{O}$, has a relative formula mass of 173.0.

Calculate a value for 'x'.

[2 marks]

Question 2b

A student completed an experiment to determine the value of 'x' in a hydrated salt.

They heated a solid sample of the salt, $\text{CoSO}_4 \cdot x\text{H}_2\text{O}$, in a crucible for 60 seconds and recorded the following set of results.

Table 2.1

Mass of empty crucible (g)	29.22
Mass of crucible and solid before heating (g)	32.67
Mass of crucible and solid after heating (g)	32.02

Use the data obtained by the student in Table 2.1 to calculate a value for 'x'.

[4 marks]

Question 2c

The actual value for x in $\text{CoSO}_4 \cdot x\text{H}_2\text{O}$ is 6.

i)
Suggest **one** reason for the difference between your calculated value in **part (b)** and the actual value.

[1]

ii)
Explain how you could improve the experiment the student did, using the same equipment, to achieve a value closer to the correct value for x.

[2]

[3 marks]

Question 2d

Cobalt(II) sulfate undergoes a displacement reaction with magnesium because magnesium is a more reactive metal than cobalt.

Write an **ionic** equation for this reaction. State symbols are **not** required.

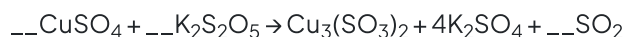
[1 mark]

Question 3a

Chevreul's salt or copper(I, II) sulfite dihydrate, $\text{Cu}_3(\text{SO}_3)_2 \cdot 2\text{H}_2\text{O}$, is an unusual hydrated salt as it contains copper in both of its common oxidation states.

The first step in the formation of Chevreul's salt is the reaction of aqueous copper sulfate with a solution of potassium metabisulfite, $\text{K}_2\text{S}_2\text{O}_5$.

Balance the equation for the reaction to form the anhydrous Chevreul's salt.



[2 marks]

Question 3b

The chemical formula of Chevreul's salt is sometimes written as $\text{Cu}_2\text{SO}_3 \cdot \text{CuSO}_3 \cdot 2\text{H}_2\text{O}$ to show that it contains copper(I) sulfite and copper(II) sulfite.

Calculate the percentage by mass of copper(I) ions in copper(I) sulfite

Show your working.

[3 marks]

Question 3c

Chevreul's salt can react with ammonia solution to form a deep blue solution of the tetraamminecopper(II) complex, $[\text{Cu}(\text{NH}_3)_4]^{2+}(\text{aq})$. It can also react with hydrochloric acid to form a white solid of copper(I) chloride, water and sulfur dioxide.

Write an **ionic** equation for the formation of copper(I) chloride from Chevreul's salt. State symbols are **not** required.

[2 marks]

Question 3d

When Chevreul's salt is heated in an inert atmosphere, it is stable to around 200 °C. Above this temperature, a variety of compounds can be produced although the final product of sustained heating at high temperatures is copper(II) oxide.

During the heating process, a sample of the mixture is taken. After separation, the most common compound was analysed and found to contain 39.8% copper, 20.1% sulfur and the rest oxygen.

Calculate the empirical formula of the most common compound. Show your working.

[3 marks]

Question 4a

The reaction of nitrous oxide gas, $\text{N}_2\text{O}(\text{g})$, with carbon disulfide vapour, $\text{CS}_2(\text{g})$, is an example of a chemical luminescence reaction producing a flame with a bright blue light.



The brightness of the light produced was a reason why this reaction was once used in low-light photography.

When nitrous oxide and carbon disulfide are placed together there is no immediate reaction.

Explain how a flame can help initiate the reaction.

[1 mark]

Question 4b

A measuring cylinder containing 100 cm^3 of nitrous oxide is used in a demonstration of the reaction between nitrous oxide and carbon disulfide.

i)

Calculate the number of moles of nitrous oxide in the measuring cylinder.

[1]

ii)

Calculate the minimum number of carbon disulfide molecules required to react with the nitrous oxide at room temperature.

[2]

[3 marks]

Question 4c

In a second demonstration, 10.8 dm^3 of nitrous oxide is reacted with 5.6 dm^3 carbon disulfide vapour.

i)

Explain which reagent is in excess. Show your working.

[2]

ii)

Calculate the maximum mass of sulfur that will be produced from this demonstration.

[1]

[3 marks]

Question 4d

The demonstration is performed a third time to determine the percentage yield of sulfur of the reaction.

For this demonstration, 16.50 dm^3 of nitrous oxide is reacted with an excess of carbon disulfide at room temperature and pressure. This produces 14.32 g of sulfur.

Calculate the percentage yield for this demonstration.

[3 marks]

Question 5a

Iron compounds have a variety of uses. Many iron compounds are found in fertilisers and as food additives and supplements.

Iron sulfate is used to treat and prevent iron deficiency anaemia.

Suggest **two** possible formulae for iron sulfate.

[1 mark]

Question 5b

Iron reacts with chlorine to form iron(III) chloride.

i)

Write an equation for this reaction of iron with chlorine.

State symbols are **not** required.

[1]

ii)

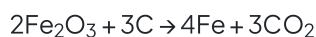
Calculate the mass of iron(III) chloride produced from 5.02 g of iron, assuming a 78% yield.

[3]

[4 marks]

Question 5c

Iron(III) oxide is reduced to Fe with carbon at a temperature of 1200 °C in the blast furnace.



An example reaction is set up by heating 9.86 g of iron(III) oxide with an excess of carbon powder. After the reaction is complete, it is left to cool to room temperature.

Calculate the volume of carbon dioxide, at room temperature, that is produced by this reduction of 9.86 g of iron(III) oxide.

[3 marks]

Question 5d

A student reacts 4.95 g of iron(III) oxide with 1.12 g of carbon.

The resulting products are 3.46 g of iron and 2.60 g of carbon monoxide.

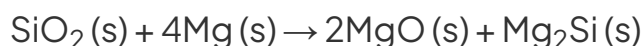
Determine the balanced equation for the reaction. Show your working.

[3 marks]

Model Answers: Medium

1a

i) The equation for the reaction of silica with magnesium is:



- Correct species; [1 mark]
- Correct balancing

AND

Correct state symbols; [1 mark]

ii) To determine the limiting reagent in this reaction:

- M_r of silica = 60.1

AND

Moles of silica = 0.00832 **OR** 8.32×10^{-3} ; [1 mark]

- Moles of magnesium = 0.0988 **OR** 9.88×10^{-2}

AND

Silica is the limiting reagent; [1 mark]

[Total: 4 marks]

- For part (i)
 - The question gives you enough information to start with a word equation
Silica + magnesium $\rightarrow$ magnesium oxide + magnesium silicide
 - **Careful:** The quartz sand is a distraction that does not form part of the equation
 - You are expected to know the formula of silica (SiO_2), magnesium (Mg) and magnesium oxide (MgO) as well as the fact that they are all solids
 - You are also given a clue about the state of silica and magnesium as they are described as powders
 - You are given the formula and state of magnesium silicide (Mg_2Si)
 - This gives you the unbalanced chemical equation
 $\text{SiO}_2(\text{s}) + \text{Mg}(\text{s}) \rightarrow \text{MgO}(\text{s}) + \text{Mg}_2\text{Si}(\text{s})$
 - The two oxygen atoms in SiO_2 will form 2MgO



- There is a total of four magnesium atoms on the right-hand side



- For part (ii)
 - The limiting reagent is the reactant with the fewest moles
 - This requires you to:
 - Calculate the M_r of silica = $28.1 + (2 \times 16.0) = 60.1$
 - The moles of silica = $\frac{0.5}{60.1} = 8.32 \times 10^{-3}$ to 3 significant figures
 - The moles of magnesium = $\frac{2.4}{24.3} = 9.88 \times 10^{-2}$ to 3 significant figures
 - Since there are fewer moles of silica, this is the limiting reagent

1b

The type of reaction is:

- Type of reaction = oxidation

OR

Combustion; [1 mark]

The chemical equation, including state symbols, is:



- Correct species; [1 mark]
- Correct balancing

AND

Correct state symbols; [1 mark]

[Total: 3 marks]

- You are told that the silane gas ignites spontaneously in air
 - This means that it is reacting with oxygen, which is a combustion / oxidation reaction
- Therefore, the chemical equation starts with:
 $\text{SiH}_4(\text{g}) + \text{O}_2(\text{g}) \rightarrow$
- The silicon will react to form silicon dioxide (silica) and the hydrogen will react to form water – this is analogous to the combustion of methane
 $\text{SiH}_4(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{SiO}_2(\text{s}) + \text{H}_2\text{O}(\text{l})$

- The silicon atom of silane will form one silicon dioxide molecule
The four hydrogen atoms of silane will form two water molecules
 $\text{SiH}_4(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{SiO}_2(\text{s}) + 2\text{H}_2\text{O}(\text{l})$
- Forming one silicon dioxide and two water molecules will require two oxygen molecules
 $\text{SiH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{SiO}_2(\text{s}) + 2\text{H}_2\text{O}(\text{l})$

1c

To calculate the maximum mass of trichlorosilane produced from the reaction of 35.125 g of silicon:

- Moles of silicon = 1.25; [1 mark]
- M_r of trichlorosilane = 135.6; [1 mark]
- Mass of trichlorosilane = 169.5 (g); [1 mark]

[Total: 3 marks]

- This is a standard reacting mass calculation
- First, calculate the moles of silicon
$$\frac{35.125}{28.1} = 1.25 \text{ moles}$$
- Since the ratio of Si : HSiCl_3 is 1 : 1, there will also be 1.25 moles of trichlorosilane
- You then calculate the M_r of trichlorosilane
 $1.0 + 28.1 + (3 \times 35.5) = 135.6$
- This allows you to calculate the mass of trichlorosilane
 $\text{Mass} = \text{moles} \times M_r = 1.25 \times 135.6 = 169.5 \text{ g}$
- Examiners often make this type of question more challenging by including percentage yields

1d

The volume of hydrogen gas required to produce 10 dm^3 of silane gas is:

- $20 (\text{dm}^3)$; [1 mark]
- (Because,) the ratio of $\text{H}_2(\text{g})$: $\text{SiH}_4(\text{g})$ is 6 : 3
OR
(Because,) the ratio of $\text{H}_2(\text{g})$: $\text{SiH}_4(\text{g})$ is 2 : 1
OR

(Because,) there are twice as many moles of hydrogen (than SiH_4 / silane); [1 mark]

[Total: 2 marks]

- You do not have to perform any complicated calculations for this question
 - The command word is **deduce**, not calculate
- One mole of any gas occupies 24 dm^3 at room temperature
- There are six moles of hydrogen and three moles of silane, i.e. twice as much hydrogen
- This means that the volume of hydrogen will be $2 \times 10 \text{ dm}^3 = 20 \text{ dm}^3$

2a

To calculate the value of x in the hydrated salt, $\text{CoSO}_4 \cdot x\text{H}_2\text{O}$:

- M_r of $\text{CoSO}_4 = 155.0$

AND

Mass of water in the hydrated salt = 18.0; [1 mark]

- Therefore, $x = 1$

OR

Therefore, the hydrated salt is $\text{CoSO}_4 \cdot \text{H}_2\text{O}$ / $\text{CoSO}_4 \cdot 1\text{H}_2\text{O}$; [1 mark]

[Total: 2 marks]

- The hydrated salt is formed from the anhydrous salt and water
 - $\text{CoSO}_4 + x\text{H}_2\text{O} \rightarrow \text{CoSO}_4 \cdot x\text{H}_2\text{O}$
- This chemical equation can be turned into a mathematical expression
 - The M_r of $\text{CoSO}_4 = 58.9 + 32.1 + (4 \times 16.0) = 155.0$
 - The M_r of $\text{H}_2\text{O} = (2 \times 1.0) + 16.0 = 18.0$
 - The question tells you the relative formula mass of the $\text{CoSO}_4 \cdot x\text{H}_2\text{O}$, 173.0
 - $155.0 + 18.0x \rightarrow 173.0$
- This rearranges to give $18.0x = 18$, so $x = 1$

2b

To calculate the value of x in the hydrated salt, $\text{CoSO}_4 \cdot x\text{H}_2\text{O}$:

- Mass of anhydrous $\text{CoSO}_4 = (32.02 - 29.22) = 2.80 \text{ g}$

AND

Mass of water removed = 0.65 g; [1 mark]

- Moles of $\text{CoSO}_4 = \frac{2.80}{155.0}$

AND

$$\text{Moles of water} = \frac{0.65}{18.0}; [1 \text{ mark}]$$

- Moles of $\text{CoSO}_4 = 0.01806\dots$

AND

$$\text{Moles of water} = 0.0361\dots; [1 \text{ mark}]$$

- Ratio of $\text{CoSO}_4 : \text{H}_2\text{O} = 1 : 1.998\dots$

AND

$$x = 2; [1 \text{ mark}]$$

OR alternative method:

- Mass of anhydrous $\text{CoSO}_4 = (32.02 - 29.22 =) 2.80 \text{ g}$

AND

$$\text{Mass of hydrated } \text{CoSO}_4 (\text{CoSO}_4 \cdot x\text{H}_2\text{O}) = (32.67 - 29.22 =) 3.45 \text{ g}; [1 \text{ mark}]$$

- Moles of $\text{CoSO}_4 = \frac{2.80}{155.0}$

AND

$$\text{Moles of } \text{CoSO}_4 = 0.01806\dots; [1 \text{ mark}]$$

- Therefore, M_r of hydrated $\text{CoSO}_4 (\text{CoSO}_4 \cdot x\text{H}_2\text{O}) = \frac{3.45}{0.01806\dots} = 190.9$

AND

$$M_r \text{ of } x\text{H}_2\text{O} = (190.9 - 155.0 =) 35.98\dots; [1 \text{ mark}]$$

- $x = \frac{35.98\dots}{18.0} = 2; [1 \text{ mark}]$

[Total: 4 marks]

- This question is a variation of the method for empirical formula

Ratio	$\frac{0.01806}{0.01806} = 1$	$\frac{0.0361}{0.01806} = 2$
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- Make sure you know the key terms:
 - Hydrated salt = the salt with water of crystallisation
 - Anhydrous salt = the salt when the waters of crystallisation have been removed through heating and there is no longer water attached
- **Careful:** A common mistake is to get the hydrated and anhydrous salts the wrong way around when working out their mass
 - The crucible with the salt before heating contains the hydrated salt
 - The crucible with the salt after heating contains the anhydrous salt

2c

i) **One** reason for the difference between the calculated value in **part (b)** and the actual value is:

Any **one** of the following:

- Failure to remove / drive off all of the water; [1 mark]
- Failure to heat for long enough; [1 mark]
- Not heated to a constant mass; [1 mark]

ii) The experiment could be improved by:

- Heating to constant mass
OR
Heating for longer
OR
Using a smaller mass; [1 mark]
- To make sure all / more of the water has been removed / driven off; [1 mark]

[Total: 3 marks]

- This is a common question for this practical
- The most common answer is to heat the solid to a constant mass

2d

The **ionic** equation for the displacement reaction between cobalt(II) sulfate and magnesium is:

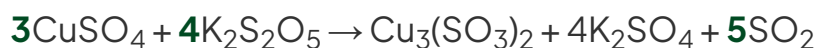
- $\text{Co}^{2+} + \text{Mg} \rightarrow \text{Mg}^{2+} + \text{Co}$; [1 mark]

[Total: 1 mark]

- The overall chemical equation for this reaction is
 - $\text{CoSO}_4 + \text{Mg} \rightarrow \text{MgSO}_4 + \text{Co}$
- **Remember:** Ionic equations should not include spectator ions
 - The sulfate ion is present in the reactants and products, so it is a spectator ion
- Removing the sulfate ion from the overall chemical equation gives the ionic equation:
 - $\text{Co}^{2+} + \text{Mg} \rightarrow \text{Mg}^{2+} + \text{Co}$

3a

The balanced equation for the formation of Chevreul's salt is:



- 3CuSO_4 ; [1 mark]
- $4\text{K}_2\text{S}_2\text{O}_5$

AND

5SO_2 ; [1 mark]

[Total: 2 marks]

- The unbalanced equation given to you as a starting point is:
 - $__\text{CuSO}_4 + __\text{K}_2\text{S}_2\text{O}_5 \rightarrow \text{Cu}_3(\text{SO}_3)_2 + 4\text{K}_2\text{SO}_4 + __\text{SO}_2$
- The formula of Chevreul's salt, $\text{Cu}_3(\text{SO}_3)_2$, suggests that 3CuSO_4 are required:
 - $3\text{CuSO}_4 + __\text{K}_2\text{S}_2\text{O}_5 \rightarrow \text{Cu}_3(\text{SO}_3)_2 + 4\text{K}_2\text{SO}_4 + __\text{SO}_2$
- The $4\text{K}_2\text{SO}_4$ formed as a product must require $4\text{K}_2\text{S}_2\text{O}_5$
 - $3\text{CuSO}_4 + 4\text{K}_2\text{S}_2\text{O}_5 \rightarrow \text{Cu}_3(\text{SO}_3)_2 + 4\text{K}_2\text{SO}_4 + __\text{SO}_2$
- This leads to a total of 11 sulfur atoms on the reactants side
 - There are 6 confirmed sulfur atoms on the right-hand side in the $\text{Cu}_3(\text{SO}_3)_2$ and $4\text{K}_2\text{SO}_4$
 - So, 5SO_2 are required to balance the sulfur atoms
 - $3\text{CuSO}_4 + 4\text{K}_2\text{S}_2\text{O}_5 \rightarrow \text{Cu}_3(\text{SO}_3)_2 + 4\text{K}_2\text{SO}_4 + 5\text{SO}_2$

3b

The percentage by mass of copper(I) ions in copper(I) sulfite is:

- $M_r \text{Cu}_2\text{SO}_3 = 207.1$; [1 mark]
- Percentage of Cu(I) ions in $\text{Cu}_2\text{SO}_3 = \frac{(2 \times 63.5)}{207.1} \times 100$; [1 mark]
- Percentage of Cu(I) ions in $\text{Cu}_2\text{SO}_3 = 61.3\%$; [1 mark]

[Total: 3 marks]

- **Careful:** Copper(I) sulfite is Cu_2SO_3
- You need to use the equation:
 - $\text{Percentage by mass} = \frac{A_r \times \text{number of atoms or ions in the compound}}{M_r \text{ of the compound}} \times 100$
- The M_r of copper(I) sulfite = $(2 \times 63.5) + 32.1 + (3 \times 16.0) = 207.1$
- So, the percentage by mass of copper(I) ions is $\frac{(2 \times 63.5)}{207.1} \times 100 = 61.3\%$

3c

The **ionic** equation for the reaction of Chevreul's salt with hydrochloric acid to form copper(I) chloride is:



- Correct species; [1 mark]
- Correct balancing; [1 mark]

[Total: 2 marks]

- **Remember:** In part (a), you are told that Chevreul's salt contains copper(I) sulfite and copper(II) sulfite
 - This means that copper(I) sulfite can be responsible for one of the reactions stated and copper(II) sulfite responsible for the other reaction
- The question states that Chevreul's salt reacts with hydrochloric acid to form copper(I) chloride, water and sulfur dioxide
 - In this case, it is copper(I) sulfite reacting with hydrochloric acid to form copper(I) chloride, water and sulfur dioxide
 - $\text{Cu}_2\text{SO}_3 + 2\text{HCl} \rightarrow 2\text{CuCl} + \text{SO}_2 + \text{H}_2\text{O}$
- **Careful:** You are being asked for the **ionic** equation for the formation of copper(I) chloride
 - This means that you need the copper(I) and the chloride ions
 $\text{Cu}^+ + \text{Cl}^- \rightarrow \text{CuCl}$
 - But, copper(I) sulfite contains two copper(I) ions

- Therefore, the equation needs to be doubled



3d

To calculate the empirical formula of the most common compound:

- Percentage of oxygen = 40.1%; [1 mark]
- Moles of copper = 0.6268

AND

Moles of sulfur = 0.6262

AND

Moles of oxygen = 2.506; [1 mark]

- Ratio of copper : sulfur : oxygen is 1 : 1 : 4

AND

Therefore, the empirical formula is CuSO_4 ; [1 mark]

[Total: 3 marks]

- The important information in this question is that the most common compound contains 39.8% copper, 20.1% sulfur and the rest oxygen
 - The amount of oxygen can be calculated by $100 - 39.8 - 20.1 = 40.1\%$
- The remainder of the question is a standard empirical formula calculation

Element	Cu	S	O
Value from question	39.8	20.1	40.1
A_r	63.5	32.1	16.0
Moles	$\frac{39.8}{63.5} = 0.6268$	$\frac{20.1}{32.1} = 0.6262$	$\frac{40.1}{16.0} = 2.506$
Ratio	$\frac{0.6268}{0.6262} = 1.001$	$\frac{0.6262}{0.6262} = 1$	$\frac{2.506}{0.6262} = 4.002$

- Therefore, the empirical formula is CuSO_4

4a

A flame helps initiate the reaction by:

- Providing the particles with the required activation energy

OR

Increasing the energy of some particles so that it is greater than the activation energy; [1 mark]

[Total: 1 mark]

- The reaction between nitrous oxide and carbon disulfide is similar to the reaction of magnesium with oxygen
- It needs a small amount of energy to make some of the particles have energy greater than the activation energy
- Once the reaction has started, the flame produced by the reaction provides sufficient energy for the remaining particles to react

4b

i) The number of moles of nitrous oxide is:

- 0.00417
OR
 4.17×10^{-3} (moles); [1 mark]

ii) The minimum number of carbon disulfide molecules required to react with the nitrous oxide at room temperature is:

- Number of moles of carbon disulfide = 0.002083
OR
 2.083×10^{-3} (moles); [1 mark]
- Minimum number of molecules of carbon disulfide = 1.25×10^{21} ; [1 mark]

[Total: 3 marks]

- For part (i)
 - **Remember:** At room temperature (20 °C) and pressure (1 atm) one mole of any

- 0.00417

OR

4.17×10^{-3} (moles); [1 mark]

ii) The minimum number of carbon disulfide molecules required to react with the nitrous oxide at room temperature is:

- Number of moles of carbon disulfide = 0.002083

OR

2.083×10^{-3} (moles); [1 mark]

- Minimum number of molecules of carbon disulfide = 1.25×10^{21} ; [1 mark]

[Total: 3 marks]

- For part (i)
 - **Remember:** At room temperature (20 °C) and pressure (1 atm) one mole of any gas has a volume of 24.0 dm^3
 - 100 cm^3 of nitrous oxide = 0.1 dm^3
 - So, the number of moles of nitrous oxide is $\frac{0.1}{24} = 0.00417$ moles
- For part (ii)
 - Using your equation from part (a), there are half as many moles of carbon disulfide
 - $2\text{N}_2\text{O (g)} + \text{CS}_2 \text{ (g)} \rightarrow 2\text{S (s)} + 2\text{N}_2 \text{ (g)} + \text{CO}_2 \text{ (g)}$
 - So, the number of moles of carbon disulfide is $\frac{0.00417}{2} = 0.002083$ moles
 - To calculate the minimum number of carbon disulfide molecules:
 - Number of molecules = moles x Avogadro's constant
 - Number of molecules = $0.002083 \times 6.02 \times 10^{23} = 1.25 \times 10^{21}$ molecules

4c

i) The reagent which is in excess is:

- Moles of nitrous oxide = 0.450

AND

Moles of carbon disulfide = 0.233; [1 mark]

- Nitrous oxide / N_2O is in excess **AND** because there is a 2 : 1 relationship between nitrous oxide and carbon disulfide

OR

Nitrous oxide / N_2O is in excess **AND** because half as much carbon disulfide is needed

OR

Nitrous oxide / N_2O is in excess **AND** because twice as much nitrous oxide is needed; [1 mark]

ii) To calculate the maximum mass of sulfur that will be produced from this demonstration:

- Moles of sulfur = 0.450

AND

Mass of sulfur = 14.4 (g); [1 mark]

[Total: 3 marks]

- To determine which reagent is in excess, you have to:
 - Determine the number of moles of both reagents
 - Nitrous oxide = $\frac{10.8}{24.0} = 0.450$ moles
 - Carbon disulfide = $\frac{5.6}{24.0} = 0.233$ moles
 - Account for the stoichiometric relationship of the reagents
 - There is $2\text{N}_2\text{O} : \text{CS}_2$
 - This means that 0.450 moles of N_2O would require 0.225 moles of carbon disulfide / CS_2
 - This means that the carbon disulfide is in excess and the nitrous oxide is the limiting reagent
- To calculate the mass of sulfur formed:
 - Since the limiting reagent is the nitrous oxide which exists in a 1 : 1 ratio with sulfur, the number of moles of nitrous oxide = the number of moles of sulfur
 - Mass = moles $\times A_r$

4d

To calculate the percentage yield for the demonstration using 16.50 dm^3 of nitrous oxide:

- Moles of nitrous oxide = 0.6875; [1 mark]
- Theoretical yield of sulfur = 22.07 (g); [1 mark]

- Percentage yield of sulfur = 64.9%; [1 mark]

[Total: 3 marks]

- Percentage yield = $\frac{\text{actual yield}}{\text{theoretical yield}} \times 100$
- The actual yield is given in the question as 14.32 g
- To determine the theoretical yield:
 - Calculate the moles of nitrous oxide
 - $\frac{16.50}{24.0} = 0.6875$ moles
 - Determine the moles of sulfur
 - Since the ratio of nitrous oxide : sulfur is 1 : 1, the number of moles of nitrous oxide will be the same as the number of moles of sulfur, 0.6875 moles
 - Calculate the theoretical mass of sulfur produced in a 100% yield reaction
 - Mass = moles $\times$ A_r = $0.6875 \times 32.1 = 22.07$ g
- So, the percentage yield of sulfur = $\frac{14.32}{22.07} \times 100 = 64.9\%$

5a

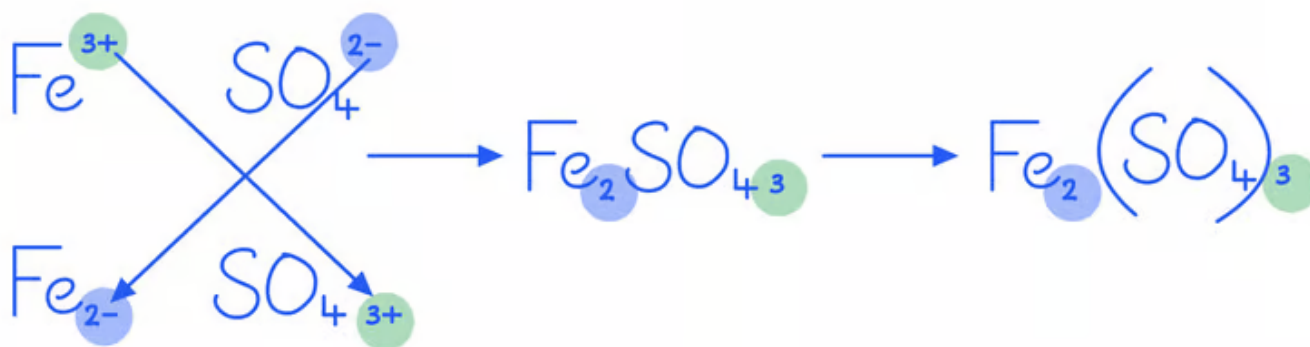
Two possible formulae of iron sulfate are:

- FeSO_4
AND
 $\text{Fe}_2(\text{SO}_4)_3$; [1 mark]

[Total: 1 mark]

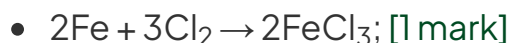
- The question asks for the formulae of iron sulfate but it does not specify if this is using iron(II) or iron(III)
- The sulfate ion has the formula SO_4^{2-}
- The iron(II) ion has the formula Fe^{2+}
 - This means that the formula for iron(II) sulfate is FeSO_4 as the two charges balance
- The iron(III) ion has the formula Fe^{3+}
 - This means that you need two Fe^{3+} ions for a total of 6+ and three SO_4^{2-} ions for a total of 6-
 - This results in the formula for iron(III) sulfate being $\text{Fe}_2(\text{SO}_4)_3$
 - You can also determine the formula of a compound using the drag-and-drop

method for the charges:



5b

i) The equation for the reaction of iron with chlorine to form iron(III) chloride is:



ii) To calculate the mass of iron(III) chloride produced from 5.02 g of iron, assuming a 78% yield:

- Moles of iron = 0.0900

AND

Moles of iron(III) chloride = 0.0900; [1 mark]

- Mass of iron(III) chloride assuming 100% yield = 14.6 (g); [1 mark]
- Mass of iron(III) chloride with a 78% yield = 11.4 (g); [1 mark]

[Total: 4 marks]

- The key to this question (and many other calculation questions) is to have a correctly balanced chemical equation to work from
- For part (i)
 - You should be able to build the unbalanced chemical equation from the information given
 - $\text{Fe} + \text{Cl}_2 \rightarrow \text{FeCl}_3$
 - With two chlorine atoms on the left-hand side and three chlorine atoms on the right-hand side, you need to use six chlorine atoms on both sides
 - $\text{Fe} + 3\text{Cl}_2 \rightarrow 2\text{FeCl}_3$
 - This means that there are now two iron atoms on the right-hand side, which is balanced by doubling the number of iron atoms on the left-hand side



- For part (ii)
 - Calculate the number of moles of iron being used
 - $\text{Moles} = \frac{\text{mass}}{A_r} = \frac{5.02}{55.8} = 0.08996\dots$ which can be rounded to 0.09
 - The stoichiometric relationship of Fe : FeCl₃ is 1 : 1
 - Therefore, there are also 0.09 moles of iron(III) chloride
 - Calculate the mass of iron(III) chloride produced from 5.02 g of iron
 - $\text{Mass} = \text{moles} \times M_r = 0.09 \times 162.3 = 14.6\dots \text{ g}$
 - This value might change depending on how many decimal places you round to as the calculator answer is 14.6011828 if you use your calculator answer to the number of moles
 - Calculate the mass of iron(III) chloride produced with a 78% yield
 - $\frac{14.6}{100} \times 78$ **OR** $14.6 \times 0.78 = 11.4 \text{ g}$

5c

To calculate the volume of carbon dioxide that is produced by the reduction of 98.6 g of iron(III) oxide:

- M_r of iron(III) oxide = 159.6
- AND**
- Moles of iron(III) oxide = 0.0618; [1 mark]
- Moles of carbon dioxide = 0.0927; [1 mark]
- Volume of carbon dioxide = 2.22 dm³; [1 mark]

[Total: 3 marks]

- You have the information required to calculate the moles of iron(III) oxide
 - M_r of iron(III) oxide = $(2 \times 55.8) + (3 \times 16.0) = 159.6$

- Calculate the number of moles of iron being used
 - $\text{Moles} = \frac{\text{mass}}{A_r} = \frac{5.02}{55.8} = 0.08996\dots$ which can be rounded to 0.09
- The stoichiometric relationship of Fe : FeCl₃ is 1 : 1
 - Therefore, there are also 0.09 moles of iron(III) chloride
- Calculate the mass of iron(III) chloride produced from 5.02 g of iron
 - $\text{Mass} = \text{moles} \times M_r = 0.09 \times 162.3 = 14.6\dots \text{ g}$
 - This value might change depending on how many decimal places you round to as the calculator answer is 14.6011828 if you use your calculator answer to the number of moles
- Calculate the mass of iron(III) chloride produced with a 78% yield
 - $\frac{14.6}{100} \times 78$ **OR** $14.6 \times 0.78 = 11.4 \text{ g}$

5c

To calculate the volume of carbon dioxide that is produced by the reduction of 98.6 g of iron(III) oxide:

- M_r of iron(III) oxide = 159.6
- AND**
- Moles of iron(III) oxide = 0.0618; [1 mark]
- Moles of carbon dioxide = 0.0927; [1 mark]
- Volume of carbon dioxide = 2.22 dm³; [1 mark]

[Total: 3 marks]

- You have the information required to calculate the moles of iron(III) oxide
 - M_r of iron(III) oxide = $(2 \times 55.8) + (3 \times 16.0) = 159.6$
 - Moles of iron(III) oxide = $\frac{\text{mass}}{M_r} = \frac{9.86}{159.6} = 0.0618$
- The ratio of Fe₂O₃ : CO₂ is 2 : 3
 - So, the number of moles of carbon dioxide = $\frac{0.0618}{2} \times 3 = 0.0927$
- One mole of any gas at room temperature and pressure occupies 24.0 dm³
 - So, 0.0927 moles of carbon dioxide will occupy $0.0927 \times 24.0 = 2.22 \text{ dm}^3$

5d

To determine the balanced equation for the reaction:

- Moles of iron(III) oxide = 0.031
AND
Moles of carbon = 0.093
AND
Moles of iron = 0.062
AND
Moles of carbon monoxide = 0.092; [1 mark]
- Ratio = $1\text{Fe}_2\text{O}_3 : 3\text{C} : 2\text{Fe} : 3\text{CO}$; [1 mark]
- Equation = $\text{Fe}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Fe} + 3\text{CO}$; [1 mark]

[Total: 3 marks]

- You cannot score all the marks for this question by simply writing and balancing the equation
- You have to calculate the number of moles of all the chemicals involved and determine the ratio of the chemicals relative to each other
 - Moles of each chemical
 - Moles of iron(III) oxide = $\frac{4.95}{159.6} = 0.031$
 - Moles of carbon = $\frac{1.12}{12.0} = 0.093$
 - Moles of iron = $\frac{3.46}{55.8} = 0.062$
 - Moles of carbon monoxide = $\frac{2.60}{28.0} = 0.092$
 - Ratio - divide by the smallest
 - Iron(III) oxide = $\frac{0.031}{0.031} = 1$
 - Carbon = $\frac{0.093}{0.031} = 3$
 - Iron = $\frac{0.062}{0.031} = 2$
 - Carbon monoxide = $\frac{0.092}{0.031} \approx 3$